

The `river.pdf` three-level meander model

Every equation integrated in Dedalus, and every formula behind the movie panels

numerical/deliverable1_noboru_model

Source: `literature/river.pdf` — 21-page deck “*Meanders of alluvial rivers as forced Rossby waves*”.

Colour key: `[river.pdf p.n]` = printed on that page of the deck. **[NOT IN DECK]** = this package’s reconstruction or choice.

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1 What the deck prints

1.1 Base state `[river.pdf p.9]`

A jet in a straight channel of half-width b , with a parabolic profile:

$$\bar{u}(y) = -\frac{d\bar{\psi}}{dy} = U_0 + \frac{\Delta}{b^2}(b^2 - y^2), \quad y \in [-b, b]. \quad (1)$$

So $\bar{u}(\pm b) = U_0$ at the banks and $\bar{u}(0) = U_0 + \Delta$ at the centre. The parabola is the load-bearing choice: $d^2\bar{u}/dy^2 = -2\Delta/b^2$ is *constant*, so the cross-channel gradient of background vorticity

$$\frac{\partial \bar{\zeta}}{\partial y} = -\frac{d^2\bar{u}}{dy^2} = \frac{2\Delta}{b^2} = \text{const} \quad (2)$$

is uniform. That constant is the channel analogue of a planetary β , and it is what makes the word “Rossby” in the deck’s title mean something.

1.2 Three levels [\[river.pdf p.9\]](#)

The deck does not carry a y -grid. It samples the channel at exactly three lines $y_j = +b, 0, -b$ and writes

$$\psi_j(x, t) = \bar{\psi}(y_j) + \hat{\psi}_j e^{i(kx - \omega t)}, \quad j = 1, 2, 3. \quad (3)$$

The cross-channel Laplacian becomes the 3-point stencil on that stencil of lines:

$$\boxed{\zeta'_2 = \frac{\psi'_1 + \psi'_3 - 2\psi'_2}{b^2} - k^2 \psi'_2} \quad \text{[river.pdf p.9]} \quad (4)$$

1.3 Interior vorticity equation [\[river.pdf p.10\]](#)

$$\boxed{\left[\frac{\partial}{\partial t} + (U_0 + \Delta) \frac{\partial}{\partial x} \right] \zeta'_2 + \frac{2\Delta}{b^2} \frac{\partial \psi'_2}{\partial x} = -C_f \frac{U_0 + \Delta}{H} \zeta'_2} \quad (5)$$

Left: advection by the centreline speed, plus the β -term $(2\Delta/b^2)\partial_x \psi'_2$. Right: **bed friction** — a Rayleigh drag on the *vorticity*, with C_f the bottom drag coefficient and H the depth.

1.4 Erodeable banks [\[river.pdf p.19\]](#)

$$\boxed{\frac{\partial \psi'_1}{\partial t} = \frac{\varepsilon C_f U_0}{b} (\psi'_2 - \psi'_1)} \quad (6)$$

The deck prints this for the $+b$ bank only; the $-b$ bank obeys the same law, which the $y \rightarrow -y$ symmetry of the base state requires. **[NOT IN DECK]** for the ψ'_3 copy.

Friction and erosion are different equations. This is worth stating explicitly because both carry C_f and are easy to conflate:

	bed friction	bank erosion
acts on	interior vorticity ζ'_2	bank streamfunctions ψ'_1, ψ'_3
lives in	a term inside (5)	the separate equation (6)
rate	$C_f(U_0 + \Delta)/H$	$\varepsilon C_f U_0/b$
extra factor	—	erodibility ε
speed used	$U_0 + \Delta$ (centre)	U_0 (bank)
role	<i>dissipation</i> : damps the disturbance	<i>relaxation</i> : pulls the bank toward the interior

Only (6) closes the feedback loop. Pages 12–18 hold the banks rigid and pose a *forced* problem; page 19 releases them and the problem becomes unstable.

2 Non-dimensionalisation

Lengths in b , speeds in $U_0 + \Delta$, time in $b/(U_0 + \Delta)$.

The time unit is not free. The deck never states it, but it is pinned by the deck's own two statements: under $t \rightarrow tb/(U_0 + \Delta)$ the friction coefficient of (5) becomes $C_f(U_0 + \Delta)T/H$, and this equals the sidebar's $\gamma = C_f b/H$ *only* for $T = b/(U_0 + \Delta)$. Then [\[river.pdf p.12–20\]](#)

$$\boxed{k^* = kb, \quad D = \frac{\Delta}{U_0 + \Delta}, \quad \gamma = \frac{C_f b}{H}, \quad E = \varepsilon C_f \frac{U_0}{U_0 + \Delta} = \varepsilon C_f (1 - D)} \quad (7)$$

and the base jet is $\bar{u}^* = 1 - Dy^2$, so $\bar{\psi}^*(y) = -(y - Dy^3/3)$ and $|\bar{\psi}^*(\pm b)| = 1 - D/3$.

Four numbers, not nine. $b, H, U_0, \Delta, C_f, \varepsilon$ never appear individually anywhere in the integration — only through $\{k^*, D, \gamma, E\}$. The package’s CONFIG therefore exposes exactly those.

3 What Dedalus integrates (1-D)

The deck is entirely normal-mode and steady-state; it contains no time integration. This package supplies one. **[NOT IN DECK]** for the periodic domain, N_x, dt, t_{end} and Dedalus itself.

Two changes of form, both exact for the single harmonic the runs contain: $-k^2\psi'_2 \rightarrow \partial_x^2\psi'_2$ in (4), and ∂_x for ik . The state is $(\psi'_1, \psi'_2, \psi'_3)$ on one periodic x -basis — the three “levels” are the y -direction, so there is no second dimension. Hence 1-D.

$$\zeta'_2 = \frac{\psi'_1 + \psi'_3 - 2\psi'_2}{b^2} + \partial_x^2\psi'_2 \quad \text{[river.pdf p.9], operator form [NOT IN DECK]} \quad (8)$$

$$\partial_t\zeta'_2 + \partial_x\zeta'_2 + 2D\partial_x\psi'_2 + \gamma\zeta'_2 = 0 \quad \text{[river.pdf p.10], nondimensional} \quad (9)$$

$$\partial_t\psi'_1 + E\psi'_1 - E\psi'_2 = 0, \quad \partial_t\psi'_3 + E\psi'_3 - E\psi'_2 = 0 \quad \text{[river.pdf p.19]} \quad (10)$$

ζ'_2 is a *linear combination of the state*, so `dt(zeta2)` is handed to Dedalus directly and it builds the mass matrix itself — no hand-elimination of $\partial_t\psi'_1, \partial_t\psi'_3$ is needed. (An earlier version did eliminate them by hand into six terms; the two forms agree to 1.4×10^{-15} , and the unexpanded one is the one a reader recognises as p.10.)

Every term is linear and treated implicitly (SBDF2), so dt is an accuracy choice, not a CFL limit.

3.1 Initial condition: the forced steady state

Not a free choice. Pages 12–16 and 18 are titled “Forced steady state” / “Forced-dissipative steady state”, p.11 states the closure as $\hat{\psi}_2 = f\hat{\psi}_1$, and pp.17–18 show the same thing in foam and dye: a *rigid carved wavy channel* with water running through it. So the initial-value problem the deck actually poses is *take that steady state, then switch the banks from rigid to erodible and watch*.

p.11 never writes f . Setting $W = 0$ (steady) in the p.10 balance with $\hat{\psi}_1 = \hat{\psi}_3$:

$$(ik^* + \gamma)[2\hat{\psi}_1 - (2 + k^{*2})\hat{\psi}_2] + 2iDk^*\hat{\psi}_2 = 0 \implies f = \frac{\hat{\psi}_2}{\hat{\psi}_1} = \frac{2(ik^* + \gamma)}{(2 + k^{*2})(ik^* + \gamma) - 2iDk^*} \quad (11)$$

[NOT IN DECK] for the algebra; the *statement* $\hat{\psi}_2 = f\hat{\psi}_1$ is p.11. At $\gamma = 0$ this collapses to $f = 2/(2 + k^{*2} - 2D)$, which is exactly the p.14 box: $|\hat{\psi}_2| > |\hat{\psi}_1|$ iff $k^{*2} < 2D$. The run then starts from

$$\psi'_1 = \psi'_3 = A_0 \cos(k^*x), \quad \psi'_2 = \text{Re}[f A_0 e^{ik^*x}]. \quad (12)$$

This state is *not* an eigenmode of the p.19 problem, so the dynamics still has to find the growing mode rather than be handed it.

A free null test. Set $E = 0$ (i.e. $\varepsilon C_f = 0$): the banks freeze, and since ψ'_2 already sits at its forced steady value nothing whatever evolves. Measured: $\max |\psi_j(t) - \psi_j(0)| \sim 10^{-16}$, growth rate 0.000000. That confirms both that the initial condition really is the p.12–18 steady state and that erosion is the sole engine.

4 Dispersion relation

p.19 prints only “ $M(\omega)[\hat{\psi}_1; \hat{\psi}_2] = 0 \Rightarrow \det M = 0$ ”. The entries are never written out, so the following is **[NOT IN DECK]** — derived here by eliminating $\hat{\psi}_1$ between the bank law and the centre balance. With $W = -i\omega^*$:

$$\text{centre [river.pdf p.9]+[river.pdf p.10]: } (W + ik^* + \gamma)[2\hat{\psi}_1 - (2 + k^{*2})\hat{\psi}_2] + 2iDk^*\hat{\psi}_2 = 0 \quad (13)$$

$$\text{bank [river.pdf p.19]: } (W + E)\hat{\psi}_1 = E\hat{\psi}_2 \quad (14)$$

Eliminating $\hat{\psi}_1$ gives a quadratic in W :

$$(2 + k^{*2})W^2 + A_1W + A_0 = 0, \quad \begin{aligned} A_1 &= (2 + k^{*2})(ik^* + \gamma + E) - 2iDk^* - 2E \\ A_0 &= E[k^{*2}(ik^* + \gamma) - 2iDk^*] \end{aligned} \quad (15)$$

and $\omega^* = iW$. The growth rate and phase speed are

$$\sigma = \text{Im } \omega^*, \quad c = \frac{\text{Re } \omega^*}{k^*}. \quad (16)$$

03_verify.py asserts (15) against the Dedalus IVP; they agree to seven decimals.

Branch continuation matters. Near $k^* \sim \gamma$ the two roots approach each other, and picking “the one with larger $\text{Im } \omega$ ” jumps between branches. `bank_branch` therefore starts from the analytic small- k^* limit $W \simeq iEDk^*/\gamma$ and tracks by proximity.

Both σ and c scale with E . Hence with the assumed εC_f . The growth *band*, the sign of c , the $\hat{\psi}_2/\hat{\psi}_1$ ratios and the k^* of the peak do not. ε is never defined and never given a value in the deck — it appears only inside the product $\varepsilon C_f U_0/b$ — so `eps_Cf=0.5` is an **assumption**, not a citation and not a fit.

5 The movie panels

Four rows. **No normalisation anywhere:** the curves are $\psi_j = \bar{\psi}(y_j) + \psi'_j(x, t)$ in their own units, with fixed axis limits for the whole movie, so what appears to grow is what actually grows. The vertical separation of the three curves *is* $\bar{\psi}(y_j)$, computed from the printed $\bar{u}$ — not an invented offset.

5.1 Row 1 — the three streamfunctions

ψ_1, ψ_2, ψ_3 against x , offset by $\bar{\psi}^*(y_j) = -(y_j - Dy_j^3/3)$, i.e. $\mp(1 - D/3)$ at the banks and 0 at the centre. ψ_1 and ψ_3 are the channel walls; ψ_2 is the interior response.

5.2 Row 2 — banks coloured by the local momentum flux

p.16 prints an *x-averaged* statement:

$$\frac{\partial}{\partial y} \langle u'v' \rangle \simeq -\langle v'_2 \zeta'_2 \rangle = \frac{D\gamma}{2b} \langle \zeta'^2_2 \rangle > 0 \quad [\text{river.pdf p.16}] \quad (17)$$

Row 2 drops the average and colours each bank by the *local* eddy vorticity flux, whose x -mean is exactly the deck’s quantity:

$$q(x, t) = -v'_2(x, t) \zeta'_2(x, t), \quad \text{top bank } +bq, \quad \text{bottom bank } -bq \quad (18)$$

The opposite signs are not a choice: $u'v'$ is odd in y for the sinuous mode, which is what the arrows on pp.16/18/19 assert. Both factors are spectral on the periodic grid:

$$v'_2 = \partial_x \psi'_2 \quad [\text{NOT IN DECK}], \quad \zeta'_2 = \frac{\psi'_1 + \psi'_3 - 2\psi'_2}{b^2} + \partial_x^2 \psi'_2. \quad (19)$$

v'_2 is used by p.16 but *never defined* in the deck; $v' = \partial_x \psi'$ is the standard reading and is what is used here. No u' at the banks is ever needed or invented — the deck gives no expression for it.

Colour scale. SymLogNorm with linthresh = max $|f|/60$, symmetric about zero. The flux is *quadratic* in the amplitude, so it spans twice as many decades as the meander does; on a linear scale every early frame would be flat.

A discrepancy in the deck, recorded not hidden. The right-hand side of (17) is printed with coefficient $D\gamma/2b$, but carrying the steady-state algebra of the p.10 balance through gives $\gamma/2D$. These are not the same number. `momentum_flux()` returns both the middle expression $-\langle v'_2 \zeta'_2 \rangle$ and the deck's right-hand side and plots them together, but `03_verify.py` asserts only the *sign* — because the sign is all the deck actually asserts.

5.3 Row 3 — growth of $|\hat{\psi}_2|$

The measured amplitude against the analytic rate. σ and c are fitted from the late-time record:

$$\sigma = \frac{d}{dt} \ln |\hat{\psi}_2|, \quad c = -\frac{1}{k^*} \frac{d}{dt} \arg \hat{\psi}_2, \quad (20)$$

both over $t \geq 0.6 t_{\text{end}}$. The fit also returns the maximum deviation of $\ln |\hat{\psi}_2|$ from the straight line over that window: if the subdominant root has not yet died the fit is curved and the residual says so, rather than the caller silently trusting a blended rate. The dashed reference line is $|\hat{\psi}_2|(0)e^{\sigma_a t}$ with σ_a from (15), so the two can be compared by eye.

5.4 Row 4 — the dispersion relation [river.pdf p.20]

$\sigma(k^*)$ and $c(k^*)$ along the bank branch of (15), with this run's k^* marked. The two headline runs sit on opposite sides of the resonance condition:

$k^* = 0.30$	$k^{*2} = 0.09 < 2D = 1.0$	resonant: grows, marches <i>upstream</i> [river.pdf p.12, top]
$k^* = 1.50$	$k^{*2} = 2.25 > 2D = 1.0$	non-resonant: decays, $\sim$ stationary [river.pdf p.12, bottom]

6 What is *not* from the deck

- The time integration itself: periodic domain, N_x , dt , t_{end} , Dedalus. All 21 pages are normal-mode or steady-state.
- A numerical value for ε (§4).
- The operator forms ∂_x^2 , ∂_x in place of $-k^2$, ik (equivalent for one mode, more general here).
- The explicit entries of $M(\omega)$ and the quadratic (15); p.19 prints only $\det M = 0$.
- The closure f of (11); p.11 states that $\hat{\psi}_2 = f\hat{\psi}_1$ but never writes f .
- The definition $v' = \partial_x \psi'$ (§5.2).

- The ψ'_3 bank equation (symmetry-required, but printed only for ψ'_1).

The initial condition is *not* on this list: it is the deck's own pp.12–18 forced steady state, which is the state the flume of pp.17–18 is physically in before the banks are released.

Provenance. Notation table and page-by-page mapping: `docs/lit_review.md`. Verification gate: `postprocessing/03_verify.py`, which runs before any figure script and aborts the build if the model no longer reproduces the deck.